

(X) A fixed model of alphabets is used with probabilities given below.

| Symbol      | a   | e   | i   | o   | u   | !   |
|-------------|-----|-----|-----|-----|-----|-----|
| Probability | 0.2 | 0.3 | 0.1 | 0.2 | 0.1 | 0.1 |

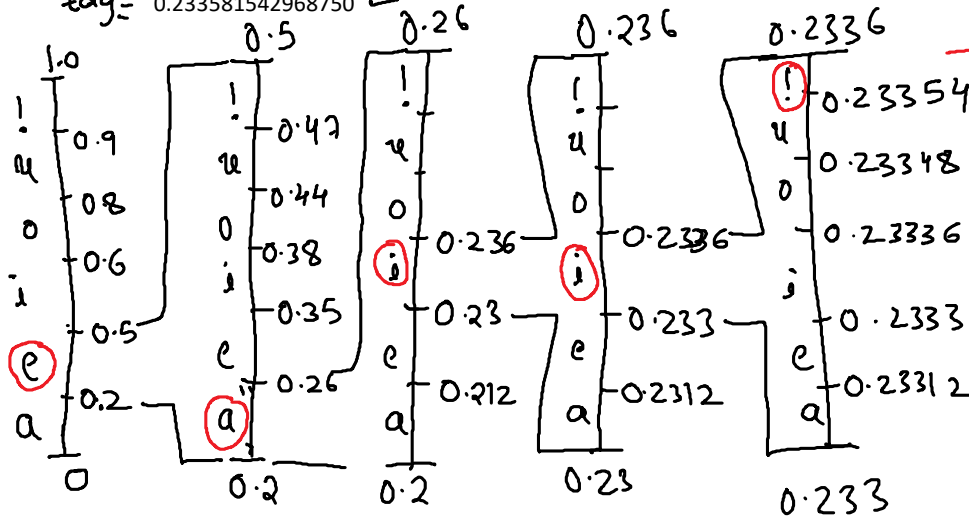
Consider '!' as the end-of-symbol. Imagine both the encoder and decoder knows the source statistics. Decode the symbols if the received binary codeword is '00111011110011'

(Sol:) Recd Binary code word = 00111011110011

$$\text{tag} = (0.00111011110011)_2 = ( ? )_{10}$$

$$\text{tag} = 0 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3} + 1 \times 2^{-4} + 1 \times 2^{-5} + 0 \times 2^{-6} + 1 \times 2^{-7} + 1 \times 2^{-8} + 1 \times 2^{-9} + 1 \times 2^{-10} + 0 \times 2^{-11} + 0 \times 2^{-12} + 1 \times 2^{-13} + 1 \times 2^{-14}$$

$$\text{tag} = 0.233581542968750$$



The decoded data is

"eaii!"

$$\begin{array}{l} 0.2 + 0.3 \times 0.2 = 0.26 \\ 0.2 + 0.3 \times 0.5 = 0.35 \\ 0.2 + 0.3 \times 0.6 = 0.38 \\ 0.2 + 0.3 \times 0.8 = 0.44 \\ 0.2 + 0.3 \times 0.9 = 0.47 \end{array} \quad \begin{array}{l} 0.2 + 0.06 \times 0.2 = 0.212 \\ 0.23 + 0.006 \times 0.2 = 0.2312 \\ 0.233 + 0.0006 \times 0.2 = 0.23312 \end{array}$$

(\*) Decode the received binary code word 'i' <sup>upto 3 symbols</sup> using arithmetic coding with the following source statistics.

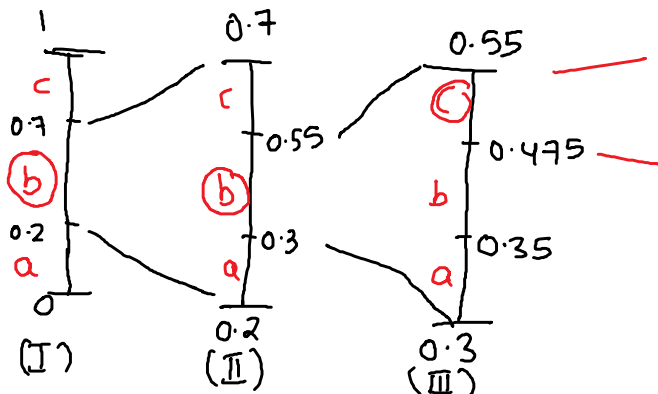
$$p\{a, b, c\} = \{0.2, 0.5, 0.3\}$$

(Sol) The received binary code word = 'i'

$$\text{tag} = (\text{0. ~~xxxx~~ binary code word}) = (0.1)_2 = 1 \times 2^{-1} = (0.5)_{10}$$

$$\boxed{\text{tag} = 0.5}$$

$$\begin{aligned} & (0.101)_2 \\ &= 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3} \\ &= (0.625)_{10} \end{aligned}$$



$$\begin{aligned} & \text{(I)} \\ & 0.2 + 0.5 \times 0.2 = 0.3 \\ & 0.2 + 0.5 \times 0.7 = 0.55 \\ & \text{(II)} \\ & 0.3 + 0.25 \times 0.2 = 0.35 \\ & 0.3 + 0.25 \times 0.7 = 0.475 \end{aligned}$$

(I) The tag value = 0.5, lies in between  $[0.2, 0.7)$ . Hence the first decoded symbol is 'b'

(II) tag value lies in between  $[0.3, 0.55)$ . Hence the second decoded symbol is 'b'

(III) Tag value lies in b/w  $[0.475, 0.55)$ . Hence the 3rd decoded symbol is 'c'

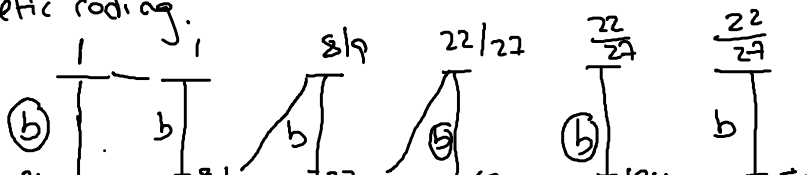
The decoded symbol for the received binary code word 'i' is bbc

Note: After decoding "bbc", 'c' lies in  $[0.475, 0.55)$ .

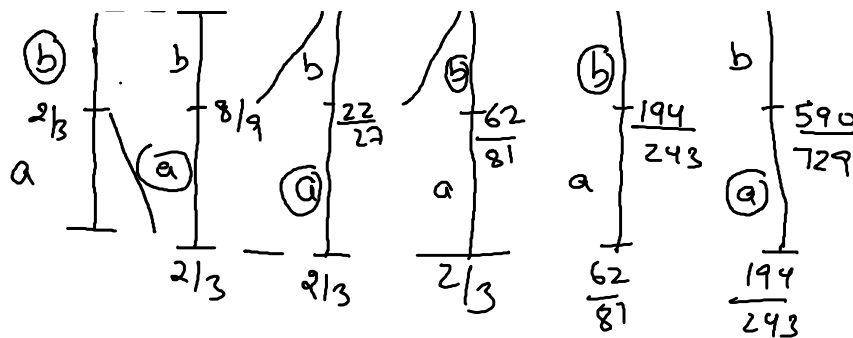
$$h-l = 0.55 - 0.475 = 0.075$$

$$p(\text{bbc}) = p(b)p(b)p(c) = 0.5 \times 0.5 \times 0.3 = 0.075 = (h-l)$$

$\{a, b\} = \{\frac{2}{3}, \frac{1}{3}\}$ . find the transmitted <sup>binary</sup> code word for baabba using arithmetic coding. tag interval is



$$\left[ \frac{194}{243}, \frac{590}{729} \right)$$



$$\left( \frac{1}{243}, \frac{1}{729} \right)$$

$$\frac{22}{27} - \frac{62}{81} = \frac{66-62}{81} = \frac{4}{81}$$

$$\frac{62}{81} + \frac{4}{81} \times \frac{2}{3} = \frac{126+8}{243} = \frac{194}{243}$$

$$\frac{2}{3} + \frac{4}{27} \times \frac{2}{3} = \frac{4}{3} \times \frac{31}{27} - \frac{62}{81}$$

$$\frac{22}{27} - \frac{194}{243} = \frac{4}{243}; \quad \frac{194}{243} + \frac{4}{243} \times \frac{2}{3}$$

$$\frac{194}{243} \times 2 = 1 + \frac{145}{243}$$

$$\frac{590}{729} \times 2 = 1 + \frac{451}{729}$$

$$\frac{145}{243} \times 2 = 1 + \frac{47}{243}$$

$$\frac{451}{729} \times 2 = 1 + \frac{173}{729}$$

$$\frac{47}{243} \times 2 = 0 + \frac{94}{243}$$

$$\frac{173}{729} \times 2 = 0 + \frac{346}{729}$$

$$\frac{94}{243} \times 2 = 0 + \frac{188}{243}$$

$$\frac{346}{729} \times 2 = 0 + \frac{692}{729}$$

$$\frac{188}{243} \times 2 = 1 + \frac{133}{243}$$

$$\frac{692}{729} \times 2 = 1 + \frac{655}{729}$$

$$\frac{133}{243} \times 2 = 1 + \frac{23}{243}$$

$$\frac{655}{729} \times 2 = 1 + \frac{581}{729}$$

$$\frac{23}{243} \times 2 = 0 + \frac{46}{243}$$

$$\frac{581}{729} \times 2 = 1 + \frac{433}{729}$$

Different

The fixed code word is "1100111".

Huffman Coding:  $I \rightarrow H(X)$

$$H(x) \leq L \leq H(x) + 1$$

To improve efficiency in of the code, we are going for extension of DMS

$$\mu(x^n) \leq 2_{n-1} \mu(x^n) + 1$$

$$n H(x) \leq \bar{L}_n \leq n H(x) + 1$$

$$H(x) \leq \frac{1}{5} \leq H(x)_{H,1}$$

If  $n \rightarrow \infty$  then efficiency is high.

$S = \{s_1, s_2, \dots, s_k\} \rightarrow$  16 no of symbols

After going for extension of DMS upto 'n'

totally how many symbols  $\rightarrow (K^n) \rightarrow \text{high}$   
 $\rightarrow \text{prob}$

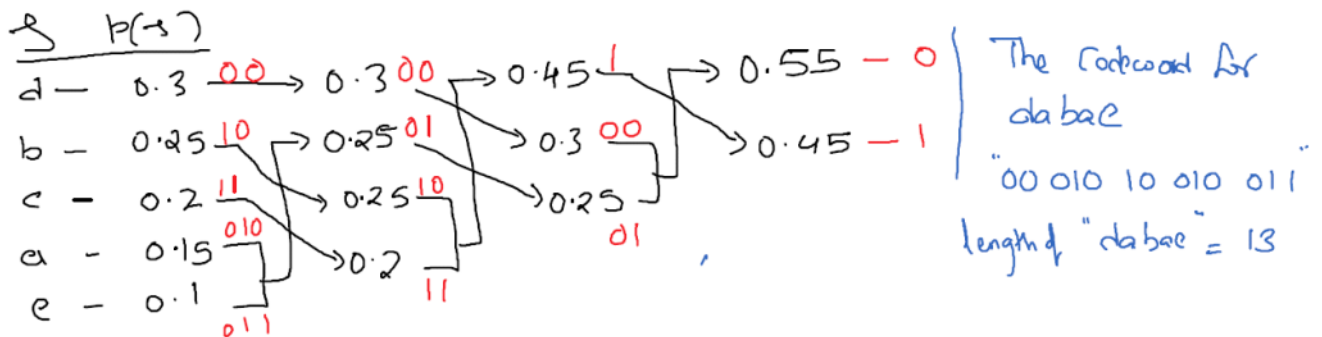
To address this drawback we are going for Arithmetic Coding

$$l_i^v = \log_2 \frac{1}{p_i}$$

- Consider the message symbols originating from DMS with the following probability distribution. Encode the data "dabae".

|                      |      |      |     |     |     |
|----------------------|------|------|-----|-----|-----|
| Symbol $\mathcal{S}$ | a    | b    | c   | d   | e   |
| $P(\mathcal{S})$     | 0.15 | 0.25 | 0.2 | 0.3 | 0.1 |

(Sol:-) First, letting  $n=1$ , will see Huffman coding



- Consider the message symbols originating from DMS with the following probability distribution. Encode the data "dabae" using Arithmetic Coding.

| Symbol $S$ | a    | b    | c   | d   | e   |
|------------|------|------|-----|-----|-----|
| $P(S)$     | 0.15 | 0.25 | 0.2 | 0.3 | 0.1 |

(Sol:) In arithmetic coding, <sup>(lexicographic)</sup> dictionary order of symbols are considered.

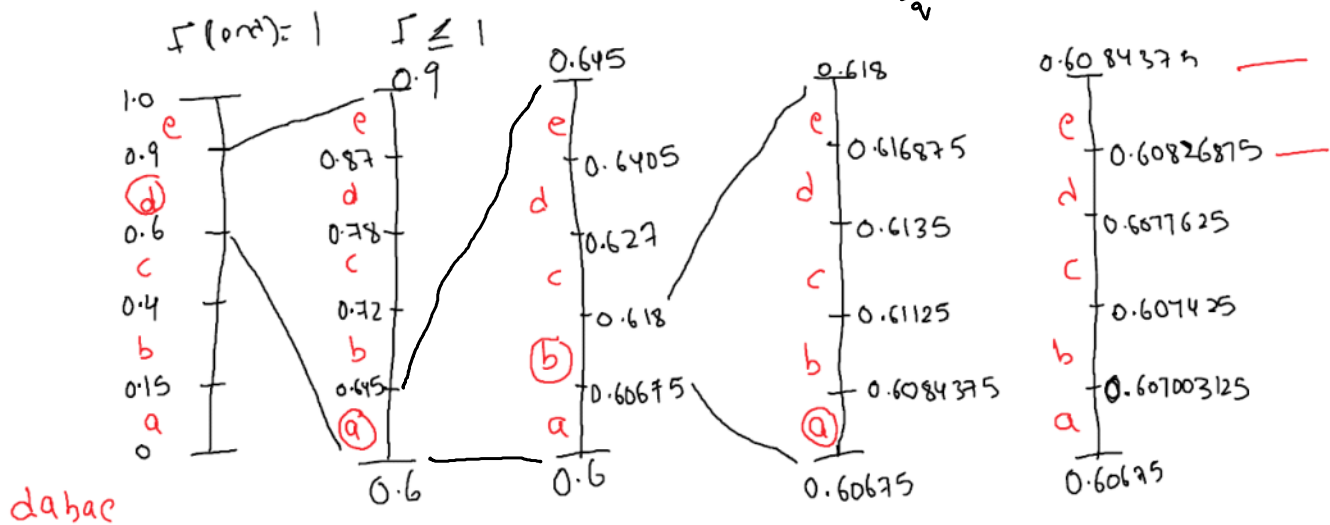
$$P: [0.15, 0.25, 0.2, 0.3, 0.1]$$

find cumulative Prob Distribution

$$F = [0.15, 0.15+0.25, 0.15+0.25+0.2, \dots] =$$

$$F = [0.15, 0.4, 0.6, 0.9, 1]$$

$$p/a \quad (1/3) \neq 0.333$$

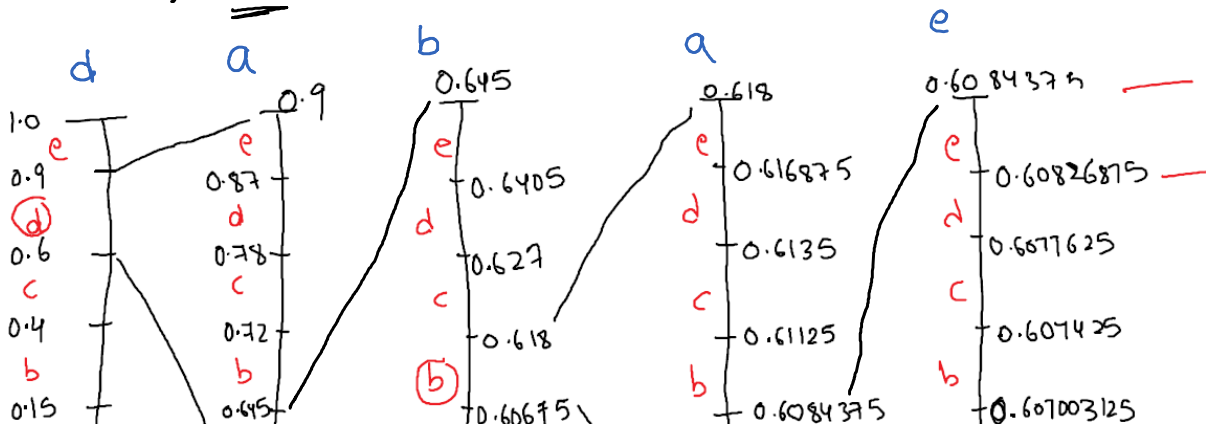


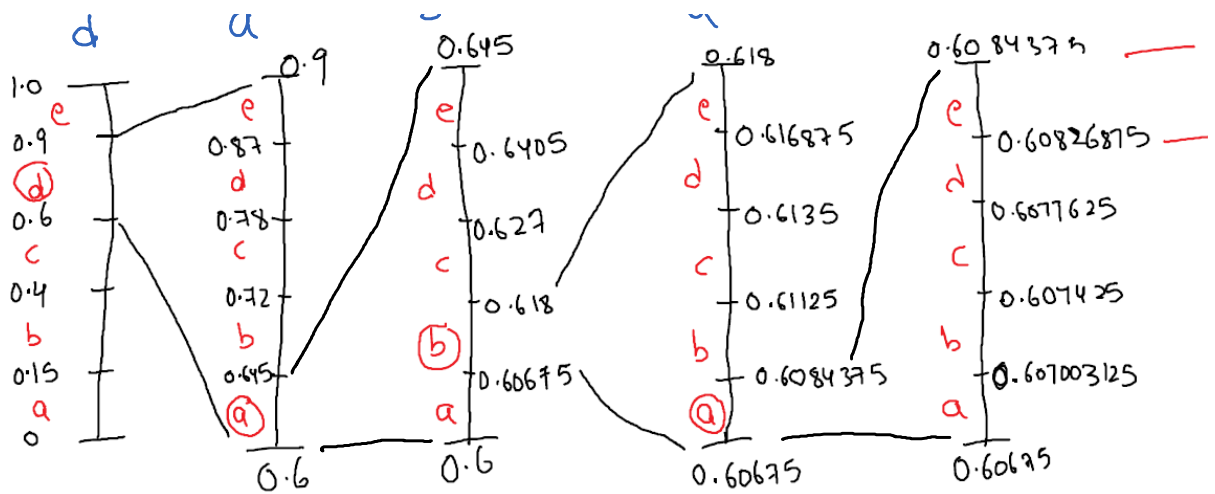
$$\begin{array}{l} 0.6 + 0.3 \times 0.15 = 0.645 \\ 0.6 + 0.3 \times 0.4 = 0.72 \\ 0.6 + 0.3 \times 0.6 = 0.78 \\ 0.6 + 0.3 \times 0.9 = 0.87 \end{array} \quad \begin{array}{l} 0.6 + 0.045 \times 0.15 = 0.60675 \\ 0.6 + 0.045 \times 0.4 = 0.618 \\ 0.6 + 0.045 \times 0.6 = 0.627 \\ 0.6 + 0.045 \times 0.9 = 0.6405 \end{array} \quad \begin{array}{l} 0.60675 + 0.01125 \times 0.15 = \\ 0.60675 + 0.01125 \times 0.4 = \\ 0.60675 + 0.01125 \times 0.6 = \\ 0.60675 + 0.01125 \times 0.9 = \end{array}$$

$$e \in \rightarrow [0.60826875, 0.6084375] \rightarrow \text{tag interval}$$

We need to select a tag value in the above tag interval

$$\text{tag} = \underline{0.6083} \rightarrow \text{transmitted to the receiver}$$





There are 2 conditions to specify stopping criteria for decoding

\* End-of-symbol.

In the above example, consider 'e' as the end-of-symbol.

fixed

\* No. of symbols.

In the above example, if we fix the no. of symbols upto 5 then decode upto 5 symbol.

\* How the tag value converted to binary form.

$$\text{tag} = 0.6083 = ( )_2$$

$$\text{tag interval } [0.60826875, 0.6084375)$$

$$0.60826875 \times 2 = 1 + 0.2165375$$

$$0.6084375 \times 2 = 1 + 0.216875$$

$$0.2165375 \times 2 = 0 + 0.433075$$

$$0.216875 \times 2 = 0 + 0.43375$$

$$0.433075 \times 2 = 0 + 0.86615$$

$$0.43375 \times 2 = 0 + 0.8675$$

$$0.86615 \times 2 = 1 + 0.7323$$

$$0.8675 \times 2 = 1 + 0.735$$

$$0.7323 \times 2 = 1 + 0.4646$$

$$0.735 \times 2 = 1 + 0.47$$

$$0.4646 \times 2 = 0 + 0.9292$$

$$0.47 \times 2 = 0 + 0.94$$

$$0.9292 \times 2 = 1 + 0.8584$$

$$0.94 \times 2 = 1 + 0.88$$

$$0.8584 \times 2 = 1 + 0.7168$$

$$0.88 \times 2 = 1 + 0.76$$

$$0.7168 \times 2 = 1 + 0.4336$$

$$0.76 \times 2 = 1 + 0.52$$

$$0.4336 \times 2 = 0 + 0.8672$$

$$0.52 \times 2 = 1 + 0.04$$

$$\text{tag} = (0.100101111)_2 = 0.6083984375$$

The no. of bits are 10

The codeword using A.E to transfer data is

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1001101111  $\rightarrow$  whose length is 10.